



BMS COLLEGE OF ENGINEERING

DEPARTMENT OF MACHINE LEARNING

LAB CYCLE – 2026 EVEN

VII Semester CIE

COURSE: MLOps	Date: 04th May 2026
COURSE CODE: 24AM6AEMLO	TOTAL: 30 Marks

Demonstrate a working solution on your laptop or on a lab system

Question Paper Code: mlops-cie-166	USN: 1BM23AI192
---	------------------------

This question paper is unique to you — do not share.

Instructions

1. You may use any **AI tools**, the internet, and any reference material
2. Use **random_state=42** and **test_size=0.2** for all train/test splits
3. Use MLflow for experiment tracking (local file store is fine)
4. All results must be **machine-readable JSON** — no screenshots
5. Do not modify the provided CSV data
6. The **results/*.json** files are your proof of execution — without them, you score 0 for that task

Submission

1. Create a GitHub repository named exactly: **Internals_Basics**
2. Inside it, create a folder named exactly: **MLOPs_Lab_CIE**
3. All your code, data, models, and results go inside MLOPs_Lab_CIE/
4. Make the repository **PUBLIC** before the deadline
5. Repo name and folder name must match **EXACTLY**

Scenario

You are an MLOps Engineer at **LaunchPredict**. LaunchPredict provides launch window optimization. The flight safety team needs to predict countdown hold duration from weather and payload parameters.

Your dataset has these features:

- **wind_speed_kmph** — (0–40)
- **humidity_pct** — (30–90)
- **payload_mass_kg** — (500–20000)
- **vehicle_type_index** — (1–4)

Target variable: **countdown_hold_min** (minutes)

A second dataset **data/new_data.csv** contains recent records with shifted distributions.

Tasks to be Performed

Complete these 4 tasks in sequence. Each builds on the previous one.

1. Experiment Tracking & Model Comparison (6 marks)
2. Docker Packaging (8 marks)
3. FastAPI Serving (8 marks)
4. Prediction Logging & Monitoring (8 marks)

Expected Repository Structure

Internals_Basics/	
MLOPs_Lab_CIE/	
data/	
training_data.csv	<i>provided (do not modify)</i>
new_data.csv	<i>provided (do not modify)</i>
src/	
train.py	<i>training + MLflow tracking</i>
predict_cli.py	<i>CLI predictor</i>
api.py	<i>FastAPI service</i>
simulate_traffic.py	<i>send requests to API</i>
monitor.py	<i>drift detection</i>
models/	<i>saved model files</i>
logs/	
predictions.jsonl	<i>generated by API</i>
results/	
step1_s1.json	<i>Task 1 output</i>
step2_s3.json	<i>Task 2 output</i>
step3_s4.json	<i>Task 3 output</i>
step4_s5.json	<i>Task 4 output</i>
requirements.txt	
Dockerfile	<i>container definition</i>
.gitignore	

Dataset: training_data.csv

Copy the data below into **data/training_data.csv**:

```
wind_speed_kmph,humidity_pct,payload_mass_kg,vehicle_type_index,countdown_hold_min
19.9,56.7,15670.8,2,49.32
12.8,54.9,7462.7,4,78.64
26.2,74.8,19789.5,4,80.3
18.6,51.4,1730.5,1,45.06
31.0,44.3,15027.5,4,74.27
24.5,47.0,7801.2,3,66.32
35.4,88.5,15746.6,2,73.44
38.9,77.3,9657.6,4,94.8
26.9,79.3,11723.1,3,80.36
10.4,43.8,6938.6,2,41.16
5.8,53.7,16267.9,4,37.17
26.9,81.5,10793.9,1,64.01
32.0,76.8,13031.3,1,54.16
37.3,45.2,14511.1,3,58.87
0.5,47.4,5642.6,2,28.36
4.1,74.0,1642.6,2,49.43
26.1,74.1,16447.7,3,64.22
33.6,73.6,19435.5,3,87.0
10.5,68.7,5554.3,2,52.89
8.3,86.2,12974.3,3,61.48
39.7,71.3,18827.7,4,102.03
33.6,32.5,3884.7,3,69.62
8.4,71.9,11598.5,4,72.71
13.7,88.5,4823.0,1,57.49
32.4,51.2,15304.5,3,73.8
```

Dataset: new_data.csv

Copy the data below into **data/new_data.csv**:

```
wind_speed_kmph,humidity_pct,payload_mass_kg,vehicle_type_index,countdown_hold_min
43.2,69.9,34598.5,4,73.75
32.6,68.5,20281.2,3,86.63
79.0,72.5,29878.5,4,118.13
56.4,31.3,13304.2,4,81.68
23.6,39.5,22844.9,3,46.99
36.1,49.1,16628.2,3,70.97
67.6,58.6,22315.9,1,68.45
66.1,43.8,36824.9,1,45.7
23.6,64.6,35385.4,4,51.85
```

37.7, 88.2, 26232.8, 2, 52.04
29.6, 64.5, 25594.7, 4, 65.11
41.3, 47.5, 29787.0, 4, 81.83
67.3, 58.3, 39142.6, 1, 58.49
30.4, 62.6, 25787.6, 1, 32.54
65.1, 59.1, 10967.3, 2, 93.51
49.0, 46.0, 35155.2, 4, 78.98
31.9, 64.5, 26017.7, 2, 52.81
49.1, 64.9, 38734.0, 3, 54.73
39.4, 43.4, 10628.5, 3, 64.53
27.5, 87.8, 23181.0, 1, 32.71

Task 1 — Experiment Tracking & Model Comparison (6 marks)

Train **SVR** and **GradientBoosting** models on the training data.

Log the following to MLflow:

- All model hyperparameters as params
- **MAE, RMSE, R²** as metrics
- Tag: **priority = "high"**
- Experiment name: **"launchpredict-countdown-hold-min"**

Select the best model by **RMSE** (lower is better).

Save results to: **results/step1_s1.json**

```
{
  "experiment_name": "launchpredict-countdown-hold-min",
  "models": [
    {"name": "SVR", "mae": 0.0, "rmse": 0.0, "r2": 0.0},
    {"name": "GradientBoosting", "mae": 0.0, "rmse": 0.0, "r2": 0.0}
  ],
  "best_model": "...",
  "best_metric_name": "rmse",
  "best_metric_value": 0.0
}
```

Task 2 — Docker Packaging (8 marks)

Containerize a CLI prediction tool.

- Create **Dockerfile** with base image **python:3.10-slim**
- Create **src/predict_cli.py** that accepts features via argparse
- Build image as: **launchpredict-predictor:v1**

Test with:

```
docker run launchpredict-predictor:v1 --wind_speed_kmph 23.5 --humidity_pct 71.3 --payload_mass_kg 8268.3 --vehicle_type_index 2
```

Save results to: **results/step2_s3.json**

```
{
  "image_name": "launchpredict-predictor",
  "image_tag": "v1",
  "base_image": "python:3.10-slim",
  "test_input": {
    "wind_speed_kmph": 23.5,
    "humidity_pct": 71.3,
    "payload_mass_kg": 8268.3,
    "vehicle_type_index": 2
  },
  "prediction": 0.0
}
```

Task 3 — FastAPI Serving (8 marks)

Serve the best model from previous tasks via FastAPI.

- **POST /predict** — accepts features, returns prediction
- **GET /heartbeat** — returns {"status": "healthy", "model_loaded": true}
- Run on **port 8080**
- Pydantic validation for all input features with correct ranges
- Return HTTP 422 for invalid inputs

Test with this exact input:

```
{"wind_speed_kmph": 23.5, "humidity_pct": 71.3, "payload_mass_kg": 8268.3, "vehicle_type_index": 2}
```

Save results to: **results/step3_s4.json**

```
{
  "health_endpoint": "/heartbeat",
  "predict_endpoint": "/predict",
  "port": 8080,
  "health_response": {"status": "healthy", "model_loaded": true},
  "test_input": {"wind_speed_kmph": 23.5, "humidity_pct": 71.3, "payload_mass_kg": 8268.3, "vehicle_type_index": 1},
  "prediction": 0.0
}
```

Task 4 — Prediction Logging & Monitoring (8 marks)

Add prediction logging to your API and build a monitoring system.

- Log every prediction to **logs/predictions.jsonl**
- Each log entry: timestamp, input, prediction
- Send **40 normal** then **10 drifted** requests
- Build **src/monitor.py** that:
 - Compares live **wind_speed_kmph** mean vs training mean (threshold: shift > **9.59**)
 - Compares live **payload_mass_kg** mean vs training mean (threshold: shift > **3563.51**)

Save results to: **results/step4_s5.json**

```
{
  "total_predictions": 50,
  "mean_prediction": 0.0,
  "drift_detected": true,
  "alerts": [
    {
      "feature": "wind_speed_kmph",
      "train_mean": 22.3,
      "live_mean": 44.82,
      "shift": 22.52,
      "threshold": 9.59,
      "status": "ALERT"
    },
    {
      "feature": "payload_mass_kg",
      "train_mean": 11291.51,
      "live_mean": 26164.51,
      "shift": 14873.0,
      "threshold": 3563.51,
      "status": "ALERT"
    }
  ]
}
```